

Anthony Jirouschek

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Lead AI Detection Engineer | Building and Validating Enterprise Defenses

Cybersecurity leader with 15 years of experience building and scaling Threat Detection, Threat Hunting, and Purple Team programs across enterprise environments. Built repeatable testing frameworks that validated 600+ detection rules and designed custom Purple Team exercises executed continuously across multiple analyst teams. Now leading AI-driven detection engineering at Spectrum Security, translating deep detection expertise into scalable, automated security tooling. USAF Cyberwarfare veteran.

Core Competencies

Leadership & Strategy: Cross-Functional Program Leadership, Security Operations at Scale, Incident Response Strategy

Threat Research: Purple Team Program Development, Threat Hunting and Research, Attack Emulation & Adversary Simulation

Detection & Engineering: Detection Engineering and Validation, Threat Intelligence Operations, Security Tool Evaluation and Integration

AI Security: AI-Driven Threat Detection, Automated Security Tooling, LLM and Agent Security Assessment

Professional Experience

Spectrum Security | San Francisco, CA (Remote)

2026 - Present

Lead AI Detection Engineer

- Building, evaluating, and refining AI agents and prompts to provide enterprise-grade detections by automating rule creation, log analysis, and allow-listing across diverse customer environments and SIEMs.

Capital One Bank | McLean, VA

2017 - 2026

Cyber Threat Detection - Testing and Validation Lead

2024 - 2026

- Led development of Live Testing, automated alert testing TTPs through TTPForge, enabling repeatable attacks against the full detection pipeline (logging, alerting, case management). Automated 83 endpoint detections.
- Led the alert testing and validation program across 600+ cyber detection rules, combining offensive techniques with the detection engineering process to expand identification and coverage of novel attack variations.
- Built CORE (CTD Operations and Review Engine), an automation platform enabling the broader team to perform automated detection development, testing, and reviews reducing ~80% of time required across all audited functions.
- Engineered a synthetic log injection capability in AWS DataBricks that validates detections as code, enabling training across CSOC/IR, Threat Intelligence, and Insider Threat functions without requiring live attack execution.
- Managed a team of 2 while driving a cross-functional testing program that partners with detection engineering, threat intelligence, and SOC teams across organizational silos.
- Executed a comprehensive security assessment of Claude Code; independently identified and exploited critical vulnerabilities, directly influencing the hardening of enterprise-wide guardrails and compensating detections.

Purple Team / Testing Lead

2022 - 2024

- Built and managed a Purple Team program that delivered custom and relevant adversary simulations for CSOC analysts. This program was utilized to expand knowledge, identify gaps, track skill leveling, and enable leadership.
- Created the Trusted Cell function to track the Covert Adversary Simulation team's live engagements. Identified gaps in processes, logging, tooling, and developed proof-of-concept queries that became threat hunting plugins and production detections.
- Designed and executed a security tool evaluation program that tested proposed enterprise solutions against realistic attack scenarios, giving leadership data-driven procurement recommendations.

Cyber Chaos Engineering Team Lead

2019 - 2022

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- Expanded Cyber Chaos Engineering, uncovered large-scale data leaks, executed Domain Admin takeovers, and exploited vulnerabilities to identify risks, before adversaries could, resulted in a fortified security posture.
- Developed and managed an attack emulation program targeting the top 10 threat actors identified by Cyber Threat Intelligence, directly improving the organization's security posture against the most relevant threats.
- Drove Purple Exercises within live production systems, providing hands-on adversary simulation training to CSOC analysts. This work led to the formal recognition and expansion of the Purple Team as a dedicated team.
- Delivered Cyber domain-wide presentations on Cyber Chaos Engineering, built organizational buy-in and promoted the adoption of threat-informed testing practices.

Cyber Threat Hunt and Research

2017 - 2019

- Coordinated and addressed internal escalations, conducted in-depth terrain assessments, maintained quality control standards, and ensured a minimal false-positive rate.
- Mentored and developed an 8-analyst Hunt Team, received findings for review and guided skill development while executing strategic threat hunting operations that integrated internal intelligence with OSINT across diverse assets.
- Initiated and led a True Positive/True Negative exercise at the request of Senior Leadership who reported the findings to Board Members. Spearheaded the organization's response to the Cosmos Bank breach.
- Enhanced threat identification precision and operational effectiveness by integrating internal intelligence with Open-Source Intelligence (OSINT) for strategic hunt operations across diverse assets.

United States Air Force, San Antonio, TX

2011 - 2017

Sr. Cybersecurity Operations Manager / Cyber Warfare Operator

2012 - 2017

- Partnered to create a Hunt function at the Air Force Cyber Emergency Response Team (AFCERT). Successfully uncovered adversary operations, developed capabilities for defenders, and resulted in a rapid deconfliction process with NSA and DoD Red Teams.
- Contributed to the development of a proactive security posture for senior-level leadership by conducting extensive open and closed-source research and hunting to identify and report on cyber actors, their motives, infrastructure, and capabilities.
- Managed security operations for 245 locations, safeguarding 750K systems against evolving cyber threats, vulnerabilities, and advanced adversaries through proficient Incident Response strategies.

Education

Associates in Cyber Security (2017) | Associate's in Information Technology (2015)

Community College of the Air Force

Certifications

CISSP, GCIH, GCED, Sec+